

Guideline Title: Bronchospasm (Asthma & COPD)

Guideline Number: GUID-3203-M008

Issue date: 09/09/2014	Replaces Dept. Policy: Bronchospasm M008
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To recognize the patient with bronchospasm and to provide and maintain adequate oxygenation/ventilation and tissue perfusion. To provide appropriate treatment and expeditious transport to the nearest appropriate medical facility.
- II. **DEFINITIONS:**
 - a. COPD - Chronic obstructive pulmonary disease
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by *GUID3203-G002 – General Patient Care*.
 - b. Assess for adventitious breath sounds (wheezes, crackles), hyperresonance upon percussion, increased anterior and posterior chest diameter, nasal flaring, pursed lip breathing, accessory muscle use with breathing, and tachypnea.
 - c. Elicit a history to identify precipitating factors. Obtain a history of intubations or hospitalizations (indicators of the severity of their disease).
 - d. Place patient in a position of comfort, usually high Flower's position.
 - e. If adequate spontaneous respirations, administer oxygen to maintain saturation >93%.
 - f. If severe respiratory distress, administer high flow oxygen via non-rebreather mask (be cautious with oxygen administration in COPD patients as they may retain CO₂). Be prepared for advanced airway management per *GUID3203-PR020 Endotracheal Intubation*. If intubation is indicated, the preferred induction agent is Ketamine over Etomidate.
 - g. Trial Continuous positive airway pressure (CPAP) or Biphasic Positive Airway Pressure (Bi-PAP) per *GUID3203-PR018 Non-Invasive Positive Pressure Ventilation (NPPV)* if at sending facility.
 - h. If acute dyspnea present (with or without wheezing)
 - i. **Albuterol 2.5 mg in 3 ml of NS nebulized.** Repeat as necessary.
 - ii. **Ipratropium 0.02% 0.5 mg (2.5ml) nebulized.** May repeat for a total of three doses if patient in severe distress.
 - i. Consider steroids: **Methylprednisolone 125 mg IV.**
 - j. Consider **Magnesium Sulfate 2 grams IV over 10 minutes**

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- k. If the patient is experiencing a severe life-threatening reaction or shows signs of shock treat per *GUID3203-M004 Allergic Reaction/Anaphylaxis*. Consider **Epinephrine (1:1,000) 0.3-0.5 mg IM** in the anterior thigh.
 - i. Massage injection site vigorously for 30-60 seconds.
 - ii. Use with caution in patients with risk factors for or known cardiovascular disease. In these patients, administer Epi only in cases of severe bronchospasm and when imminent threat to life exists.
- l. Consider **Terbutaline 0.25 mg SC**, repeat in 15 minutes (max dose 0.5 mg in 4 hours).

IV. DOCUMENTATION:

- a. EMS Charts

V. REFERENCES: